Retail Customer Purchase Prediction and Product Recommendation System

Submitted By:-
Suraj Deep Kumar Mehta
IN26010558
AI/ML Internship Program

Introduction

Retail businesses generate large amounts of transaction data.

Customer purchase prediction helps improve retention and marketing.

Machine Learning identifies future purchasing behaviour.

Streamlit provides an interactive web interface.

Problem Statement

Retailers find it difficult to predict customer buying behavior. Develop an AI system that analyzes transaction history and predicts future purchases for personalized recommendations.

Objectives

- Analyze retail transaction data.
- Clean and preprocess customer records.
- Engineer customer behaviour features.
- Train a Logistic Regression model.
- Predict customer purchases and recommend products.

Technologies Used

- Python
- Pandas & NumPy
- Scikit-learn
- Streamlit
- Plotly
- Joblib

System Workflow

- Retail Dataset
- Data Cleaning & Validation
- Feature Engineering
- Logistic Regression Model
- Purchase Prediction
- Product Recommendation
- Interactive Dashboard

Machine Learning Model

- Algorithm: Logistic Regression
- Binary Classification
- Target: Purchase within prediction horizon
- Evaluation: Accuracy, Precision, Recall, F1 Score, ROC-AUC

Application Features

- Interactive dashboard
- Revenue by month and city
- Customer purchase probability
- Hybrid product recommendation
- Download recommendations as CSV

Results

- Predicts customer purchase probability.
- Provides High, Medium, and Low probability bands.
- Generates personalized product recommendations.
- Displays business insights through charts and KPIs.

Advantages & Future Scope

- Easy-to-use interface
- Automated preprocessing
- Supports retail decision making
- Future: Real-time database, cloud deployment, advanced ML, customer segmentation

Thank you